

NAN, Thailand

WMO: 483310

Lat: 18.7669N

Lon: 100.7633E

Elev: 202

StdP: 98.93

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	13.8	9.2	7.4	22.6	11.0	8.4	21.5	4.4	26.6	3.7	26.4	0.2	20	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.9	38.2	24.9	37.1	24.8	36.0	24.8	27.6	33.2	27.2	32.6	26.9	32.2	1.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	22.3	29.4	26.0	21.9	29.2	25.7	21.5	28.9	89.5	33.4	87.8	32.8	86.3	32.5	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.4	3.7	3.1	DB	9.7	40.2	2.0	1.1	8.3	41.0	7.1	41.7	6.0	42.3	4.6	43.2	(4)
(5)			WB	9.3	29.1	2.0	0.8	7.9	29.6	6.7	30.1	5.6	30.6	4.1	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	22.9	25.0	27.8	29.9	29.6	29.3	28.5	28.0	28.2	27.6	25.5	22.9	(6)	
	DBStd	2.96	2.05	2.00	2.12	1.86	1.88	1.42	1.55	1.30	1.19	1.37	2.03	2.42	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	3	2	0	0	0	0	0	0	0	0	0	0	2	(9)	
	CDD10.0	6243	400	421	552	597	609	578	573	559	545	546	464	400	(10)	
	CDD18.3	3205	143	187	294	347	351	328	314	301	295	287	214	144	(11)	
	CDH23.3	33082	1201	1869	3335	4229	4073	3787	3329	2963	2867	2619	1719	1091	(12)	
	CDH26.7	14834	460	948	1913	2391	2004	1671	1279	1018	1042	1041	687	379	(13)	
(14) Wind	WSAvg	0.6	0.4	0.5	0.7	0.7	0.7	0.8	0.7	0.6	0.5	0.4	0.4	0.4	(14)	
(15) Precipitation	PrecAvg	1307	10	14	33	92	170	165	210	276	206	108	19	6	(15)	
	PrecMax	1676	65	72	95	175	340	276	435	430	413	228	69	39	(16)	
	PrecMin	995	0	0	0	2	62	67	73	131	83	36	1	0	(17)	
	PrecStd	179	19	21	25	55	68	62	88	82	106	50	20	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	36.3	39.0	40.1	39.1	36.9	36.0	34.5	34.6	34.5	34.2	32.9	(19)	
		MCWB	22.3	22.6	24.0	25.5	26.2	26.4	26.0	26.9	26.7	25.8	24.8	23.7	(20)	
	2%	DB	32.0	35.0	37.7	38.9	37.2	35.2	34.3	33.2	33.6	33.5	33.0	31.5	(21)	
		MCWB	21.8	22.3	23.4	25.3	26.0	26.1	26.3	26.5	26.4	25.6	24.2	22.5	(22)	
	5%	DB	30.9	33.8	36.6	37.4	35.9	34.1	33.1	32.2	32.7	32.7	32.0	30.3	(23)	
		MCWB	21.2	22.0	23.3	25.3	26.1	26.1	26.1	26.2	26.1	25.3	23.8	21.9	(24)	
	10%	DB	29.6	32.4	35.2	36.0	34.3	33.0	32.0	31.2	31.7	31.8	30.8	29.0	(25)	
		MCWB	20.9	21.8	23.2	25.3	25.9	26.0	25.9	25.9	25.9	25.1	23.3	21.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	25.2	26.5	28.1	28.3	27.7	27.6	27.7	27.8	27.3	26.3	25.1	(27)	
		MCDB	30.4	32.9	35.7	36.1	34.4	33.2	33.0	32.2	32.1	31.9	31.4	30.1	(28)	
	2%	WB	23.3	24.1	25.5	27.1	27.4	27.2	27.1	27.2	27.2	26.7	25.6	23.9	(29)	
		MCDB	29.1	32.0	34.2	35.0	33.7	32.6	32.3	31.5	31.5	31.4	30.6	28.7	(30)	
	5%	WB	22.5	23.3	24.8	26.5	26.9	26.8	26.6	26.7	26.8	26.2	24.9	23.1	(31)	
		MCDB	28.1	31.0	33.3	34.4	33.2	32.0	31.4	30.8	30.9	30.7	29.4	27.9	(32)	
	10%	WB	21.7	22.6	24.1	26.0	26.5	26.5	26.2	26.4	26.4	25.7	24.3	22.3	(33)	
		MCDB	27.3	29.8	32.5	33.8	32.3	31.3	30.5	30.0	30.1	29.9	28.3	27.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	16.2	15.5	12.9	9.7	8.0	7.1	7.0	7.8	9.2	11.3	13.1	(35)	
		MCDBR	16.2	17.8	17.4	14.8	12.0	9.6	8.9	8.4	9.2	10.2	12.3	13.9	(36)	
	5% WB	MCWBR	6.8	6.6	5.7	4.4	3.7	3.2	3.0	3.2	3.4	3.7	4.6	5.7	(37)	
		MCWBR	14.3	15.8	15.2	13.1	10.4	8.6	8.0	7.8	8.4	9.3	10.8	12.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.444	0.546	0.691	0.659	0.495	0.445	0.440	0.458	0.486	0.525	0.431	0.430	(40)	
		taud	2.187	1.918	1.616	1.737	2.202	2.370	2.376	2.311	2.214	2.063	2.291	2.251	(41)	
	Ebn at Noon		830	761	669	698	813	847	852	844	819	770	839	831	(42)	
		Edh at Noon	138	190	266	237	148	124	123	132	144	163	124	126	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.46	5.12	5.33	5.78	5.28	4.75	4.37	4.32	4.56	4.60	4.63	4.26	(44)	
	RadStd		0.26	0.24	0.32	0.39	0.47	0.36	0.35	0.28	0.22	0.23	0.27	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+147	+149

Nomenclature: See separate page